Mihir Ravindra Adelkar

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Full-stack engineer with 3+ years building distributed production systems and AI-powered internal tooling across Python REST, React/TypeScript. Built production LLM/RAG tooling on Claude API + MCP + Pinecone with explicit attention to prompt design and output validation; at NeoSOFT, designed internal scaffolding adopted by 15+ engineering teams.

Education

Northeastern University — Boston, MA
Master of Science in Software Engineering Systems — *Sept 2023 – May 2025*

University of Mumbai — Mumbai, India
Bachelor of Engineering in Computer Engineering — *Aug 2018 – Jun 2021*

Technical Skills

Languages: Python, JavaScript, TypeScript, Java, C++, SQL, Shell
AI & LLMs: Claude, RAG, MCP, Prompt Engineering, Pinecone, Embeddings, Semantic Search, Llama 3, Ollama, NLP
Backend & APIs: FastAPI, Django, Flask, Node.js, REST, GraphQL, gRPC, Kafka (async / background pipelines)
Frontend & Data: React, Next.js, Redux, Tailwind, PostgreSQL, Redis, Elasticsearch, MongoDB
Cloud & Infra: AWS (EKS, Lambda, S3), GCP, Docker, Kubernetes, Terraform, Helm, GitHub Actions, Jenkins
Observability & Testing: Prometheus, Grafana, New Relic, Jest, Cypress, K6

Experience

Full-Stack Developer — Aug 2025 – Present
KeelWorks Foundation — *Boston, MA*

- Own end-to-end development of KeelCompass, an internal engagement platform; ship React + TypeScript with Node + PostgreSQL services, partnering with design to translate ambiguous, cross-team workflows into reliable tooling
- Leverage AI coding assistants and custom agents to accelerate scaffolding, review, and refactor cycles; hold AI-generated code to the same correctness, testing, and maintainability standards as hand-written production code
- Automated end-to-end deployment on AWS EKS with GitHub CI, eliminating manual release steps and unblocking same-day deploys; introduced a reusable component library and coding standards adopted across the team

Software Engineer — Jun. 2021 – May 2023
NeoSOFT Technologies — *Mumbai, India*

- Built internal engineering scaffolding adopted by 15+ teams: *mf-shell* (Webpack Module Federation for micro-frontend composition) and *Console* (multi-tenant admin panel with RBAC); partnered with platform leads across teams to translate their workflows into reusable, well-documented tooling
- Audited a 750K+ user real estate CRM (500ms avg latency); designed full architecture, flow diagrams, and migration plan; built a gRPC/Kafka microservices boilerplate; coordinated 20+ engineers over 6 months – 12 services migrated, 7x faster inter-service calls, 45%+ infra cost reduction via AWS rightsizing
- Designed Python/Django REST and GraphQL APIs backed by PostgreSQL and Redis caching for 750K+ users, cutting response times 70%; implemented Kafka-based async pipelines for long-running notification and event processing, decoupling latency-sensitive UX from heavy backend computation
- Engineered Homepage Builder CMS and multi-tenant CRM for Nesto (React, Node.js, AWS Lambda); enabled marketing to update pages independently, removing a recurring engineering bottleneck
- Architected Django analytics engine aggregating per-student classroom data on a Next.js dashboard, enabling teachers to monitor live attendance, quiz scores, and engagement with post-session reports
- Set up CI/CD with mandatory tests and New Relic APM dual dashboards (engineering metrics + stakeholder view), cutting MTTR 25 → 15 min; reached 85% Jest/Cypress test coverage; mentored 8 junior developers through code review and bi-weekly knowledge-sharing seminars

Recent Projects

CVE Intelligence System – RAG-Powered Security Assistant | *Go, Python, Kafka, Pinecone, Llama 3, AWS EKS, Terraform,*

- Built distributed microservices for real-time CVE ingestion using Kafka streaming with schema validation; ingestion fans out to long-running enrichment and embedding jobs without blocking upstream producers
- Developed a RAG pipeline with Pinecone and Llama 3, letting engineers query a large unstructured corpus in natural language instead of NVD searches - the same way that makes image and metadata corpora interactively explorable
- Deployed on AWS EKS with Terraform IaC and Istio service mesh; instrumented Prometheus/Grafana monitoring with automated alerting and Jenkins CI/CD

Healthcare Prior Authorization AI | *Python, FastAPI, Claude API, MCP, Pinecone, FHIR R4, Streamlit*

- Built an LLM-powered internal tool replacing 3 day manual insurance reviews with automated decisions in 10 sec; clinical NLP pipeline mapped unstructured notes to FHIR resources with 95%+ extraction accuracy on the test set
- Designed an MCP-based context architecture combining patient clinical data with insurance policy documents; owned prompt design, context window construction, retrieval, and output validation; system emits confidence scores with full audit trails so reviewers can triage and trust model output